EE114 - Homework 1

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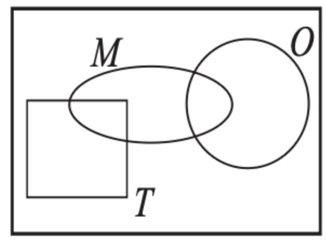
Quiz 1.1

A pizza at Gerlanda's is either regular (R) or Tuscan (T). In addition, each slice may have mushrooms (M) or onions (O) as described by the Venn diagram at right. For the sets specified below, shade the corresponding region of the Venn diagram.



(1) R
Regular pizzas (outside T)

(2) $M \cap O$
Mushrooms AND Onions (overlap)

(3) $R \cap M$
Regular AND Mushrooms (overlap M and R)

(4) $M \cup O$
Mushrooms OR Onions (union)

(5) $R \cup M$
Regular OR Mushrooms (union)

(6) $T^c - M$
Not Tuscan minus Mushrooms (Regular sans M)

Problem 1.1.1

For Gerlanda's pizza in Quiz 1.1, answer these questions:

(a) Are T and M mutually exclusive?
No. The Venn diagram shows that the set M (mushrooms) overlaps with the set T (Tuscan), meaning it is possible to have a Tuscan pizza with mushrooms. No, they are not mutually exclusive.

(b) Are R, T, and M collectively exhaustive?
Yes. Since R and T represent all possible pizza types (Regular and Tuscan) and form a partition of the space (assuming every pizza is either R or T), the union $R \cup T$ is the entire sample space. Adding M doesn't change this coverage. Yes, they are collectively exhaustive.

(c) Are T and O mutually exclusive? State this condition in words.
The diagram shows O (onions) overlaps with T (Tuscan). Thus, they are not mutually exclusive. No. Words: One can order a Tuscan pizza with onions.

(d) Does Gerlanda's make Tuscan pizzas with mushrooms and onions?
This asks if $T \cap M \cap O$ is non-empty. Based on the intersection of the M and O circles within the T region: Yes.

(e) Does Gerlanda's make regular pizzas that have neither mushrooms nor onions?
This asks if $R \cap M^c \cap O^c$ is non-empty. This corresponds to the region inside R but outside both M and O. Yes.

Problem 1.2.1

A fax transmission can take place at any of three speeds depending on the condition of the phone connection between the two fax machines. The speeds are high (h) at 14,400 b/s, medium (m) at 9600 b/s, and low (l) at 4800 b/s. In response to requests for information, a company sends either short faxes of two (t) pages, or long faxes of four (f) pages. Consider the experiment of monitoring a fax transmission and observing the transmission speed and length. An observation is a two-letter word, for example, a high-speed, two-page fax is ht.

(a) What is the sample space of the experiment?
The sample space consists of all pairs of speed $S \in \{h, m, l\}$ and length $L \in \{t, f\}$: S = {ht, hf, mt, mf, lt, lf}

(b) Let A_1 be the event "medium-speed fax." What are the outcomes in A_1?
A_1 includes all outcomes where the speed is m: $\mathbf{A_1} = \{mt, mf\}$

(c) Let A_2 be the event "short (two-page) fax." What are the outcomes in A_2?
A_2 includes all outcomes where the length is t: $\mathbf{A_2} = \{ht, mt, lt\}$

(d) Let A_3 be the event "high-speed fax or low-speed fax." What are the outcomes in A_3?
A_3 includes outcomes where speed is h or l: $\mathbf{A_3} = \{ht, hf, lt, lf\}$

(e) Are A_1, A_2, and A_3 mutually exclusive?
Check intersection $A_1 \cap A_2 = \{mt\}$. Since the intersection is not empty, they are not mutually exclusive. No.

(f) Are A_1, A_2, and A_3 collectively exhaustive?
The union $A_1 \cup A_2 \cup A_3 = \{mt, mf\} \cup \{ht, mt, lt\} \cup \{ht, hf, lt, lf\}$. Any outcome in S is present in at least one set. Yes.

Problem 1.2.2

An integrated circuit factory has three machines X, Y, and Z. Test one integrated circuit produced by each machine. Either a circuit is acceptable (a) or it fails (f). An observation is a sequence of three test results corresponding to the circuits from machines X, Y, and Z, respectively. For example, aaf is the observation that the circuits from X and Y pass the test and the circuit from Z fails the test.

(a) What are the elements of the sample space of this experiment?
Each machine has 2 outcomes (a, f). Total outcomes $2^3 = 8$. S = {aaa, aaf, afa, aff, faa, faf, ffa, fff}

(b) What are the elements of the sets

$$Z_F = \{\text{circuit from } Z \text{ fails}\},$$

$$X_A = \{\text{circuit from } X \text{ is acceptable}\}.$$

Z_F are outcomes ending in f: $\mathbf{Z_F} = \{aaf, aff, faf, fff\}$
X_A are outcomes starting with a: $\mathbf{X_A} = \{aaa, aaf, afa, aff\}$

(c) Are Z_F and X_A mutually exclusive?
The intersection $Z_F \cap X_A = \{aaf, aff\}$. Since it is not empty: No.

(d) Are Z_F and X_A collectively exhaustive?
The union $Z_F \cup X_A = \{aaa, aaf, afa, aff, faf, fff\}$. Missing outcomes $\{faa, ffa\}$. No.

(e) What are the elements of the sets

$$C = \{\text{more than one circuit acceptable}\},$$

$$D = \{\text{at least two circuits fail}\}.$$

$C = \{$2 or 3 a's$\}$: $\mathbf{C} = \{aaa, aaf, afa, faa\}$
$D = \{$2 or 3 f's$\}$: $\mathbf{D} = \{aff, faf, ffa, fff\}$

(f) Are C and D mutually exclusive?
An outcome cannot have ≥ 2 a's and ≥ 2 f's simultaneously in length 3. Yes.

(g) Are C and D collectively exhaustive?
Every outcome has either fewer than 2 f's (so ≥ 2 a's) or ≥ 2 f's. The union covers all 8 outcomes. Yes.

Problem 1.3.1

Computer programs are classified by the length of the source code and by the execution time. Programs with more than 150 lines in the source code are big (B). Programs with ≤ 150 lines are little (L). Fast programs (F) run in less than 0.1 seconds. Slow programs (W) require at least 0.1 seconds. Monitor a program executed by a computer. Observe the length of the source code and the run time. The probability model for this experiment contains the following information: $P[LF] = 0.5$, $P[BF] = 0.2$, and $P[BW] = 0.2$. What is the sample space of the experiment? Calculate the following probabilities:

Sample space is {LF, LW, BF, BW}. We can calculate $\boldsymbol{P[LW] = 1 - (0.5 + 0.2 + 0.2) = 0.1}$.

(a) $P[W]$
$\boldsymbol{P[W] = P[LW] + P[BW] = 0.1 + 0.2 = 0.3}$

(b) $P[B]$
$\boldsymbol{P[B] = P[BF] + P[BW] = 0.2 + 0.2 = 0.4}$

(c) $P[W \cup B]$
$\boldsymbol{P[W \cup B] = P[LW] + P[BW] + P[BF] = 0.1 + 0.2 + 0.2 = 0.5}$

Problem 1.3.2

There are two types of cellular phones, handheld phones (H) that you carry and mobile phones (M) that are mounted in vehicles. Phone calls can be classified by the traveling speed of the user as fast (F) or slow (W). Monitor a cellular phone call and observe the type of telephone and the speed of the user. The probability model for this experiment has the following information: $P[F] = 0.5$, $P[HF] = 0.2$, $P[MW] = 0.1$. What is the sample space of the experiment? Calculate the following probabilities:

Sample space is {HF, HW, MF, MW}. We deduce $\boldsymbol{P[MF] = P[F] - P[HF] = 0.5 - 0.2 = 0.3}$. Summing all probabilities: $\boldsymbol{P[HW] = 1 - (0.2 + 0.3 + 0.1) = 0.4}$.

(a) $P[W]$
$\boldsymbol{P[W] = P[HW] + P[MW] = 0.4 + 0.1 = 0.5}$

(b) $P[MF]$
From above: 0.3

(c) $P[H]$
$\boldsymbol{P[H] = P[HF] + P[HW] = 0.2 + 0.4 = 0.6}$

Problem 1.4.1

Mobile telephones perform *handoffs* as they move from cell to cell. During a call, a telephone either performs zero handoffs (H_0), one handoff (H_1), or more than one handoff (H_2). In addition, each call is either long (L), if it lasts more than three minutes, or brief (B). The following table describes the probabilities of the possible types of calls.

	H_0	H_1	H_2
L	0.1	0.1	0.2
B	0.4	0.1	0.1

What is the probability $P[H_0]$ that a phone makes no handoffs?
$\boldsymbol{P[H_0] = 0.1 + 0.4 = 0.5}$
What is the probability a call is brief?
$\boldsymbol{P[B] = 0.4 + 0.1 + 0.1 = 0.6}$
What is the probability a call is long or there are at least two handoffs?
$\boldsymbol{P[L \cup H_2] = P[L] + P[H_2] - P[L \cap H_2] = 0.4 + 0.3 - 0.2 = 0.5}$

Problem 1.4.2

For the telephone usage model of Example 1.14, let B_m denote the event that a call is billed for m minutes. To generate a phone bill, observe the duration of the call in integer minutes (rounding up). Charge for M minutes $M = 1, 2, 3, \ldots$ if the exact duration T is $M - 1 < t \leq M$. A more complete probability model shows that for $m = 1, 2, \ldots$ the probability of each event B_m is

$$P[B_m] = \alpha(1-\alpha)^{m-1}$$

where $\alpha = 1 - (0.57)^{1/3} = 0.171$.

(a) Classify a call as long, L, if the call lasts more than three minutes. What is $P[L]$?
$\boldsymbol{P[L] = P[M \geq 4] = \sum_{m=4}^{\infty} \alpha(1-\alpha)^{m-1} = (1-\alpha)^3 = (0.57^{1/3})^3 = 0.57}$

(b) What is the probability that a call will be billed for nine minutes or less?
$\boldsymbol{P[M \leq 9] = 1 - P[M > 9] = 1 - (1-\alpha)^9 = 1 - (0.57^{1/3})^9 = 1 - 0.57^3 \approx 0.815}$

Problem 1.4.3

The basic rules of genetics were discovered in mid-1800s by Mendel, who found that each characteristic of a pea plant, such as whether the seeds were green or yellow, is determined by two genes, one from each parent. Each gene is either dominant d or recessive r. Mendel's experiment is to select a plant and observe whether the genes are both dominant d, both recessive r, or one of each (hybrid) h. In his pea plants, Mendel found that yellow seeds were a dominant trait over green seeds. A yy pea with two yellow genes has yellow seeds; a gg pea with two recessive genes has green seeds; a hybrid gy or yg pea has yellow seeds. In one of Mendel's experiments, he started with a parental generation in which half the pea plants were yy and half the plants were gg. The two groups were crossbred so that each pea plant in the first generation was gy. In the second generation, each pea plant was equally likely to inherit a y or a g gene from each first generation parent. What is the probability $P[Y]$ that a randomly chosen pea plant in the second generation has yellow seeds?

Each parent in Gen 1 is Yg. Offspring receives Y or g with prob 0.5 from each. Outcomes for offspring genes: YY, Yg, gY, gg, each with prob 0.25. Yellow seeds occur for $\{YY, Yg, gY\}$. $P[Y] = 0.25 + 0.25 + 0.25 = 0.75$

Problem 1.4.4

Use Theorem 1.7 to prove the following facts:

(a) $P[A \cup B] \geq P[A]$
$A \cup B = A \cup (B \cap A^c)$, disjoint. $P[A \cup B] = P[A] + P[B \cap A^c]$. Since $P(\cdot) \geq 0$, proven.

(b) $P[A \cup B] \geq P[B]$
Similarly, $P[A \cup B] = P[B] + P[A \cap B^c] \geq P[B]$. Proven.

(c) $P[A \cap B] \leq P[A]$
$A = (A \cap B) \cup (A \cap B^c)$. $P[A] = P[A \cap B] + P[A \cap B^c] \geq P[A \cap B]$. Proven.

(d) $P[A \cap B] \leq P[B]$
Similarly $P[B] \geq P[A \cap B]$. Proven.

Problem 1.4.5

Use Theorem 1.7 to prove by induction the *union bound*: For any collection of events $A_1, \ldots, A_n$,

$$P[A_1 \cup A_2 \cup \cdots \cup A_n] \leq \sum_{i=1}^{n} P[A_i].$$

Base case (n=2): $\boldsymbol{P[A_1 \cup A_2] = P[A_1] + P[A_2] - P[A_1 \cap A_2] \leq P[A_1] + P[A_2]}$**.**
Inductive step: Assume for $\boldsymbol{k}$**. For** $\boldsymbol{k+1}$**, let** $\boldsymbol{B = \cup_{i=1}^{k} A_i}$**.**
$\boldsymbol{P[B \cup A_{k+1}] \leq P[B] + P[A_{k+1}] \leq (\sum_{i=1}^{k} P[A_i]) + P[A_{k+1}]}$**. Proven.**